MATTHEW S. ZHANG

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POSITIONS

C.L.E. Moore Instructor August 2026 -
MIT, Department of Mathematics, mentored by Prof. Philippe Rigollet

EDUCATION

University of Toronto January 2022 - June 2026 (expected)
PhD, Computer Science, supervised by Prof. Murat Erdogdu. Affiliated with the Vector Institute for Artificial Intelligence. GPA: 4.00/4.00

University of Toronto September 2020 - January 2022
MS, Computer Science, supervised by Profs. Murat Erdogdu and Animesh Garg. Affiliated with the Vector Institute for Artificial Intelligence. GPA: 4.00/4.00

University of Toronto September 2016 - May 2020
BASc, Engineering Science. High Honours, GPA: 3.94/4.00

VISITING POSITIONS

Yale University January 2025 - April 2026
Hosted by Prof. Sinho Chewi

Georgia Institute of Technology October 2023 - December 2023
Hosted by Prof. Santosh Vempala

ENS Paris-Saclay May 2022 - July 2022
Hosted by Prof. Alain Durmus

JOURNAL PUBLICATIONS

[1] **Uniform-in-N log-Sobolev inequality for the mean-field Langevin dynamics with convex energy**
Sinho Chewi, Atsushi Nitanda, **Matthew S. Zhang** SIMA, 2026

[2] **Analysis of Langevin Monte Carlo from Poincaré to Log-Sobolev**
Sinho Chewi, Murat A. Erdogdu, Mufan (Bill) Li, Ruoqi Shen, **Matthew S. Zhang** FoCM, 2024

CONFERENCE PUBLICATIONS

[3] **Covariance estimation with Markov chain Monte Carlo**
Yunbum Kook, **Matthew S. Zhang** ICML, 2026

[4] **Shifted Composition IV: Underdamped Langevin and Numerical Discretizations with Partial Acceleration**
Jason M. Altschuler, Sinho Chewi, **Matthew S. Zhang** STOC, 2026

[5] **Rényi-infinity constrained sampling with d^3 membership queries**
Yunbum Kook, **Matthew S. Zhang** SODA, 2025

[6] **In-and-Out: Algorithmic diffusions for sampling convex bodies**
Yunbum Kook, Santosh Vempala, **Matthew S. Zhang** NeurIPS, 2024 (spotlight)

[7] **Sampling from the mean-field stationary distribution**
Yunbum Kook, **Matthew S. Zhang**, Sinho Chewi, Murat A. Erdogdu, Mufan Li COLT, 2024

[8] **Improved discretization analysis for the underdamped Langevin Monte Carlo**
Matthew S. Zhang, Sinho Chewi, Mufan Li, Krishnakumar Balasubramanian, Murat A. Erdogdu
COLT, 2023

[9] **Tight regret and complexity bounds for Thompson Sampling via Langevin Monte Carlo**
Tom Huix, **Matthew S. Zhang**, Alain Durmus AISTATS, 2023

[10] **Towards a Theory of Non-Log-Concave Sampling: First-Order Stationarity Guarantees for Langevin Monte Carlo**
Krishnakumar Balasubramanian, Sinho Chewi, Murat A. Erdogdu, Mufan Li, Adil Salim, **Matthew S. Zhang** COLT, 2022

[11] **Convergence and Optimality of Policy Gradient Methods in Weakly Smooth Settings**
Matthew S. Zhang, Murat A. Erdogdu, Animesh Garg AAAI, 2022

[12] **Convergence of Langevin Monte Carlo in Chi-Squared and Rényi Divergence**
Murat A. Erdogdu, Rasa Hosseinzadeh, **Matthew S. Zhang** AISTATS, 2022

[13] **One-Shot Pruning of Recurrent Neural Networks by Jacobian Spectrum Evaluation**
Matthew S. Zhang, Bradly Stadie ICLR, 2020

PREPRINTS

[14] **Algorithmic warm-starts for Hamiltonian Monte Carlo**
Matthew S. Zhang, Jason Altschuler, Sinho Chewi Preprint, 2026

[15] **Sharp propagation of chaos in Rényi divergences**
Matthew S. Zhang Preprint, 2026

[16] **Stability of the Kim–Milman flow map**
Sinho Chewi, Aram-Alexandre Pooladian, **Matthew S. Zhang** Preprint, 2025

[17] **Sublinear iterations can suffice even for DDPMs**
Matthew S. Zhang, Stephen Huan, Jerry Huang, Nicholas Matthew Boffi, Sitan Chen, Sinho Chewi Preprint, 2025

[18] **Perspectives on Stochastic Localization**
Bobby Shi, Kevin Tian, **Matthew S. Zhang** Preprint, 2025

[19] **Analysis of Langevin midpoint methods using an anticipative Girsanov theorem**
Matthew S. Zhang Preprint, 2025

[20] **Benchmarking Model-Based Reinforcement Learning**
Tingwu Wang, Xuchan Bao, Ignasi Clavera, Jerrick Hoang, Yeming Wen, Eric Langlois, **Matthew S. Zhang**, Guodong Zhang, Pieter Abbeel, Jimmy Ba Preprint, 2019

INVITED TALKS AND PRESENTATIONS

Midpoint methods for SDEs: Speeding up sampling and generative modelling
Math for AI (MFAI) workshop, UNAM, Mexico City April 2026

Sublinear iterations can suffice even for DDPMs
Seminar, NUS Computer Science November 2025

Perspectives on stochastic localization
Seminar, University of Pennsylvania November 2025

Analysis of Langevin midpoint methods using an anticipative Girsanov theorem
IMS Young Mathematical Scientists Forum – Statistics and Data Science, NUS November 2025
Fast and Curious 2: MCMC in action, University of Toronto September 2025
Wasserstein Gradient Flows in Math and Machine Learning, BIRS July 2025

Toward ballistic acceleration for log-concave sampling
Seminar, NUS Mathematics — November 2025
Two Faces of Optimization, ETH — July 2025

Uniform-in-N log-Sobolev inequality for finite-particle systems
INFORMS 2025 — October 2025
Seminar, University of Tokyo — November 2024

Sampling and isoperimetry for finite particle approximations
SIAM Conference on the Mathematics of Data Science — October 2024

Sampling in the mean-field regime
Probability Summer School, Saint Flour (Student talk) — July 2024
Seminar, Yale University — March 2024

Isoperimetry and the convergence of LMC
Machine Learning Summer School, ÉMINES — July 2022

Convergence of LMC in Rényi Divergence
Applied Mathematics Seminar, CERMICS — June 2022

Analysis of LMC from Poincaré to log-Sobolev
Complexity of Sampling Working Group, Simons Institute — November 2021

AWARDS

Canada Graduate Scholarship (Doctoral) — 2023

University of Toronto Fellowships — 2021

Daisy Intelligence Scholarship for Engineering Science — 2019

Faculty of Applied Science and Engineering Award — 2018

Engineering Society Awards — 2018

Jane Elizabeth Ham Scholarship — 2017

Canadian Freshman Debating Champion — 2017

SERVICE

Conference Reviewer AISTATS (2022, 2023, 2024, 2025, 2026), NeurIPS (2022, 2023, 2024, 2025, 2026) ICLR (2023, 2025, 2026), ICML (2023, 2024, 2025, 2026), COLT (2025, 2026), ALT (2024, 2025), AAAI (2025, 2026), SODA (2026)

Journal Reviewer SPA, JAA, FoCM, JMLR, TMLR, Statistica Sinica, IMAJNA, JDS, SINUM

Teaching assistant CSC412 (Winter 2024), CSC343 (Winter 2023), CSC2532 (Winter 2022), CSC498 (Winter 2021), CSC343 (Fall 2020), ESC180 (Fall 2020)

Organized a **reading group on sampling algorithms and stochastic localization** at the Georgia Institute of Technology, 2023-2024.

REFERENCES

Murat Erdogdu, Associate Professor — erdogdu@cs.toronto.edu

Sinho Chewi, Assistant Professor — sinho.chewi@yale.edu

Daniel Lacker, Associate Professor — daniel.lacker@columbia.edu